

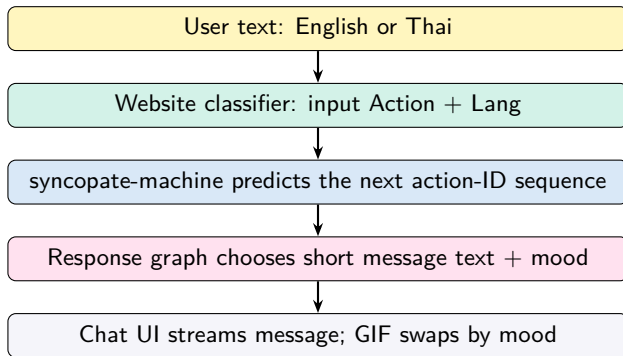
Syncopate-Machine

Tiny Browser Transformer for Action-Sequence Chat

Dancing With My Code

Action-only architecture, 2026-05-25

The New Contract



Hard rule

The model no longer predicts tokenizer pieces or raw words. It predicts **action IDs**. Text is owned by the website response graph.

Why The Tokenizer Got Fired

Old Path

- SentencePiece tokenizer
- Text-token prediction
- Browser had to decode words
- More room for weird repeats
- Bigger checkpoint and slower UI

New Path

- No tokenizer at runtime
- 23 integer action IDs
- Website owns language and message style
- Mood is explicit for animation
- Tiny checkpoint, browser-friendly

Less magic. More control. Fewer stupid browser failures.

Action Vocabulary

ID	Name	Role	ID	Name	Role
0	PAD	padding	12	Compliment	nice input
1	SOS	start	13	Agree	yes / same
2	EOS	end	14	Disagree	no / wrong
3	SEP	separator	15	General	fallback
4	Unknown	unused fallback	16	Eating	food talk
5	Greeting	hello	17	DailyLife	casual life
6	Farewell	goodbye	18	RustGo	Rust beats Go
7	Frustrated	annoyed	19	Identity	who am I
8	Sad	low mood	20	ShitTalk	roast mode
9	Happy	upbeat	21	TH	Thai
10	Question	question	22	EN	English
11	Insult	user swears			

Tiny Model Shape

Browser Model

Vocab	23
Context	64
Layers	2
Width	64
Attention heads	4
KV heads	1
FFN width	128
Params	71,424

Decoder Block

- RMSNorm
- Causal attention
- RoPE position rotation
- Grouped-query attention
- SwiGLU feed-forward
- Weight-tied output projection

The neural model decides **what kind of reply happens next**; the website decides **what words hit the chat box**.

Parameter Count

Config: $V = 23$, $N = 2$, $d = 64$, $n_h = 4$, $n_{kv} = 1$, $d_k = 16$, $d_{ff} = 128$

Component	Formula	Params
Embedding	$V \times d = 23 \times 64$	1,472
Attention per layer	$d^2 + 2dn_{kv}d_k + d^2$	10,240
FFN per layer	$3dd_{ff}$	24,576
Norms per layer	$2d$	128
Final norm	d	64
Total	$1472 + 2(10240 + 24576 + 128) + 64$	71,424

Output projection reuses the embedding matrix, so it adds zero new params.

Browser Runtime

- Website loads `model-personal-v2.mpk`
- Website loads `model-config-personal-v2.json`
- Runtime tries WebGPU first when requested
- Auto mode falls back to CPU when WebGPU is unavailable
- Chrome without a valid WebGPU adapter should not kill the chat box
- UI streams message text from graph output, not raw model tokens

Mood Contract

Model output action sequence maps to one mood: `normal`, `happy`, `frustrated`, or `sad`. The mascot GIF changes from that mood.

Training Path

Data Build

```
python examples/build_action_data.py --offline
```

CUDA Training

```
cargo run --release --features cuda --example train_action_model --  
  --steps 3000 --batch-size 32 --lr 0.003  
  --checkpoint-dir runs/action-model-personal-v2
```

- Training is CUDA-only by design.
- Website runtime can choose WebGPU or CPU.
- Training data is action sequences, not text-token corpora.

What Got Deleted

Removed Code

- ChatModel
- Trainer
- infer_tokens*
- old token generation APIs
- legacy layout engine
- old Multiscreen modules

Removed Junk

- SentencePiece experiment folder
- spm_train.exe
- dead tokenizer examples
- obsolete tests
- stray ignored nul file

Remaining crate surface: **action model + training + browser runtime.**

Deploy Shape

- Build the Leptos site with `trunk build --release`
- Ship the generated `dist/` directory to Vercel
- Keep model assets in `dancing-with-my-code-v2/assets/`
- Cache `.mpk` and `.json` assets aggressively
- Do not ship old tokenizer artifacts

Final State

The website talks like Dancing With My Code. The model only nudges conversation flow by action IDs. Clean split, less nonsense.

References

- Burn framework: <https://github.com/tracel-ai/burn>
- RoPE / RoFormer: rotary position embeddings
- GQA: grouped-query attention
- SwiGLU: gated feed-forward block
- Project docs: README.md and DEVLOG.md